Time Series Forecasting of Retail Sales

Executive Summary

This project analysed 73,100 retail observations across 5 stores and 20 products, covering January 2022 to January 2024, to understand demand patterns and develop a practical approach to forecasting daily sales. Historical sales were analysed alongside operational factors such as inventory, pricing, discounts, promotions and regional information, with several forecasting approaches evaluated before selecting the strongest-performing model.

The analysis produced the following key findings and business implications:

1. Prioritise inventory availability
Inventory should be treated as a key input to replenishment and demand planning. Inventory Level was the strongest predictor of sales in the final model and showed a clear positive relationship with Units Sold. Where commercially viable, raising minimum stock thresholds could help maintain product availability and prevent stock constraints from limiting potential sales.

2. Reduce reliance on routine discounting
Discount levels showed virtually no measurable relationship with sales. Routine discounting could therefore potentially be reduced where its primary purpose is to generate additional demand. Discounts should instead be targeted towards specific commercial objectives and evaluated against incremental sales.

3. Integrate forecasting with operational planning
The final SARIMAX forecasting model, incorporating operational and external variables alongside historical sales, substantially outperformed simpler forecasting approaches. It achieved a MAPE of 4.81% and approximately 45% lower MAE than the naive baseline.

Overall Finding
The project indicates that the greatest practical opportunity lies in linking demand forecasting more closely with inventory and supply-chain planning. Using operational information alongside historical demand can provide a stronger foundation for short-term forecasting and more informed decisions around stock availability, replenishment and commercial activity.

The analysis also found that factors such as holidays, weekends and seasonal patterns had relatively little influence on overall demand in this dataset, suggesting that day-to-day operational factors were more relevant to forecasting sales than calendar-based effects.

Table of Contents

Executive Summary **1**
1. Project Overview **4**
1.1 Business Context 4
1.2 Problem Statement 4
1.3 Objectives 5
1.4 Business Value 5
1.5 Dataset Overview 5
2. Data Understanding & Preprocessing **6**
2.1 Dataset Dimensions and Variables 6
2.2 Date Handling 7
2.3 Missing-Value Checks 7
2.4 Data-Quality Observations 7
2.5 Feature Preparation 7
2.6 Aggregation to Daily Level 8
3. Exploratory Data Analysis **8**
3.1 Sales Distribution 8
3.2 Daily/Weekly/Monthly Patterns 8
3.3 Store and Product Variation 9
3.4 Inventory vs Sales 10
3.5 Price/Discount Analysis 10
3.6 Promotion/Holiday Analysis 11
3.7 Weather and Seasonality 12
3.8 Correlation Analysis 12
4. Time-Series Analysis **13**
4.1 Daily Target Construction 13
4.2 Rolling Statistics 13
4.3 Train/Test Split 14
4.4 ACF/PACF 14
4.5 ADF Test 16
4.6 Interpretation of Stationarity 16
5. Forecasting Models **17**
5.1 Naive Baseline 17
5.2 ARIMA 17
5.3 SARIMA 17
5.4 Prophet 18
5.5 SARIMAX 18
5.6 Why the Final Model Was Selected 20
6. Business Insights and Recommendations **21**
6.1 Inventory Planning 21

6.2 Pricing..... 21
6.3 Discounts..... 22
6.4 Promotions..... 22
6.5 Regional Considerations..... 22
6.6 External-Variable Forecasting..... 23
7. Conclusion..... 23

List of Figures

Figure 1: Distribution of Units Sold 8
Figure 2: Total Daily Units Sold 9
Figure 3: Average Units Sold by Store 9
Figure 4: Inventory Level vs Units Sold 10
Figure 5: Discount vs Units Sold 11
Figure 6: Units Sold by Holiday/Promotion 11
Figure 7: Units Sold by Weather Condition 12
Figure 8: Correlation Matrix of Numerical Variables 13
Figure 9: Daily Sales with 30-Day Rolling Mean 14
Figure 10: Autocorrelation of Daily Units Sold 15
Figure 11: Partial Autocorrelation of Daily Units Sold 16
Figure 12: SARIMAX Actual vs Forecast 21

List of Tables

Table 1: Dataset Overview 5
Table 2: Dataset Variables and Descriptions 6
Table 3: Forecasting Model Performance 19

1. Project Overview

1.1 Business Context

Retail businesses operate in an environment where demand can fluctuate across time, products, stores and external conditions. Accurate demand forecasting is therefore an important component of inventory planning and supply-chain management.

When demand is underestimated, businesses may experience stockouts, lost sales and disruptions to fulfilment operations. Conversely, overestimating demand can result in excess inventory, increased holding costs and inefficient utilization of warehouse and retail space.

The project aims at using demand forecasting as a means to improve inventory planning with particular emphasis on identifying temporal patterns such as seasonality and trends, as well as incorporating external factors including promotions, holidays and weather conditions.

1.2 Problem Statement

The objective of this project is to develop a time-series forecasting approach capable of predicting future retail demand using historical sales data and relevant explanatory variables.

The analysis considers two broad sources of predictive information:

Historical demand behaviour

This includes:

- Previous sales observations
- Temporal patterns
- Trend
- Seasonality
- Autocorrelation
- Short-term fluctuations

External and operational factors

The dataset provides additional variables relating to:

- Inventory levels
- Units ordered
- Price
- Discounts
- Holiday/promotional status
- Competitor pricing

- Weather conditions
- Seasonality
- Region

The project therefore investigates both univariate time-series models, which rely primarily on historical demand, and multivariate/exogenous-variable models, which incorporate additional information alongside the target series.

1.3 Objectives

The objective of this project is to forecast future retail sales using historical demand data and relevant external factors.

Specifically, the project aims to:

- Analyse historical sales patterns, including trend, seasonality and short-term fluctuations.
- Examine whether factors such as inventory, pricing, discounts, promotions, weather and regional characteristics provide useful information for predicting demand.
- Develop and evaluate time-series forecasting models for daily retail sales.
- Compare model performance using out-of-sample forecasting metrics.
- Identify insights that can support inventory planning and supply-chain decision-making.

1.4 Business Value

Accurate demand forecasting can help retail businesses make better operational decisions by:

- Improving inventory planning and reducing the risk of stockouts or excess inventory.
- Supporting replenishment and supply-chain planning.
- Identifying external and operational factors that are associated with changes in demand.
- Providing a data-driven basis for short-term demand planning.

1.5 Dataset Overview

The project uses a retail inventory and sales dataset containing daily observations across multiple stores and products.

Table 1: Dataset Overview

Attribute	Value
Observations	73,100
Time period	1 Jan 2022 – 1 Jan 2024

Stores	5
Products	20
Variables	15
Target variable	Units Sold

Key supporting variables are Inventory Level, Units Ordered, Price, Discount, Holiday/Promotion, Competitor Pricing, Weather Condition, Seasonality and Region.

The dataset provides both historical sales information and operational/external variables, allowing demand forecasting to be approached using both univariate and exogenous-variable time-series models.

2. Data Understanding & Preprocessing

2.1 Dataset Dimensions and Variables

The dataset contains 73,100 observations and 15 variables, covering daily retail activity across 5 stores and 20 products. The variables include sales, inventory, ordering, pricing, discounts, promotions, competitor pricing, weather, seasonality and regional information.

Units Sold was selected as the primary target variable for forecasting.

Table 2: Dataset Variables and Descriptions

Variable	Type	Description
Date	Date	Daily observation date
Store ID	Categorical	Store identifier
Product ID	Categorical	Product identifier
Category	Categorical	Product category
Region	Categorical	Regional classification
Inventory Level	Numerical	Available inventory
Units Sold	Numerical	Units sold; forecasting target
Units Ordered	Numerical	Units ordered

Demand Forecast	Numerical	Existing demand forecast
Price	Numerical	Product price
Discount	Numerical	Applied discount
Weather Condition	Categorical	Recorded weather condition
Holiday/Promotion	Binary	Holiday/promotional indicator
Competitor Pricing	Numerical	Competitor price
Seasonality	Categorical	Supplied seasonal classification

2.2 Date Handling

The Date column was converted to a proper datetime format and used as the temporal index for time-series analysis. The dataset contains observations from 1 January 2022 to 1 January 2024. For time-series modelling, the forecasting period covers 1 January 2022 to 31 December 2023.

2.3 Missing-Value Checks

Missing-value checks were performed across the dataset. No missing observations requiring imputation were identified, and the daily time series contained continuous date coverage for the modelling period.

2.4 Data-Quality Observations

Basic validation was performed to ensure that the variables were suitable for analysis and forecasting. Numerical variables were reviewed for their ranges and distributions, while categorical variables were examined for their distinct values and consistency.

2.5 Feature Preparation

Categorical variables were converted into numerical representations for forecasting. Weather Condition, Seasonality and Region were aggregated to the daily level using category proportions, with one reference category omitted from each variable to avoid redundancy. A derived Price Difference feature was also created to represent the difference between competitor pricing and the product's price.

The existing Demand Forecast variable was excluded from the primary forecasting models to avoid incorporating an existing forecast as an input to the model.

2.6 Aggregation to Daily Level

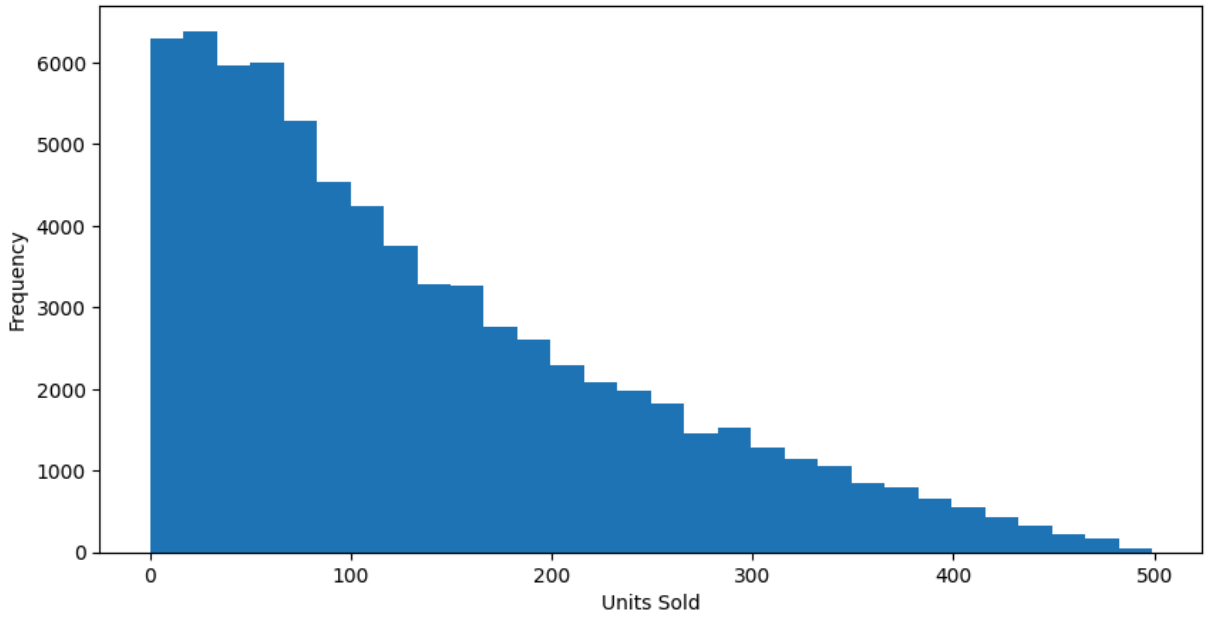
For time-series modelling, the observation-level data was aggregated into a single daily sales series by summing Units Sold across stores and products.
The external variables were similarly transformed to daily level. Numerical variables were aggregated using daily averages, while categorical variables such as weather, seasonality and region were represented using their daily proportions.
This produced a consistent dataset containing 730 daily observations for the forecasting analysis.

3. Exploratory Data Analysis

3.1 Sales Distribution

Units Sold showed a right-skewed distribution, with a mean of approximately 136 units and a median of approximately 107 units. Sales ranged from 0 to 499 units.

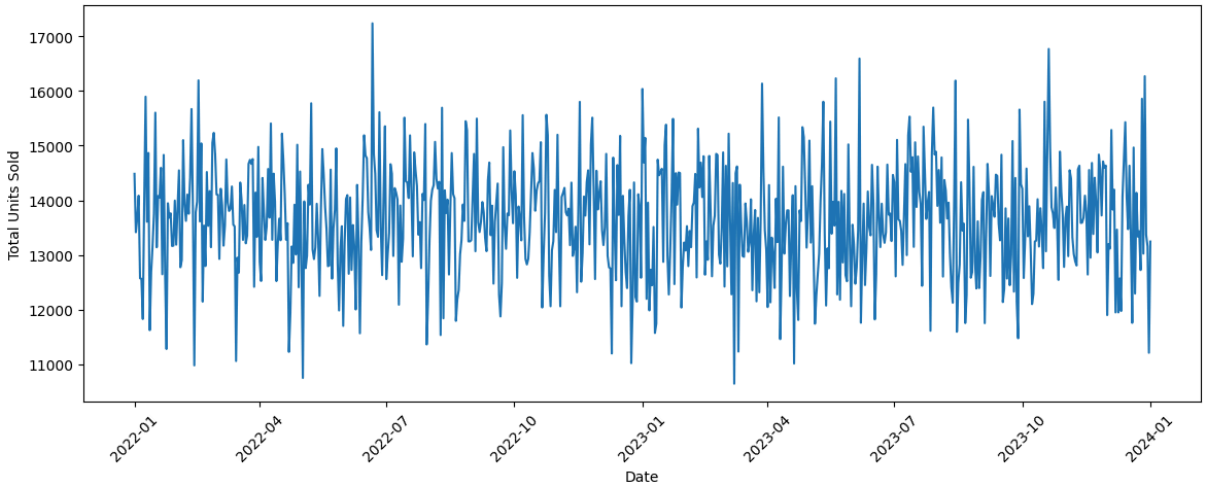
Figure 1: Distribution of Units Sold



3.2 Daily/Weekly/Monthly Patterns

Daily aggregate sales showed considerable short-term variation but no pronounced long-term trend. Weekly and monthly aggregation produced smoother patterns, with only modest differences across months and days of the week.

Figure 2: Total Daily Units Sold



Overall, the series exhibited weak temporal and seasonal structure.

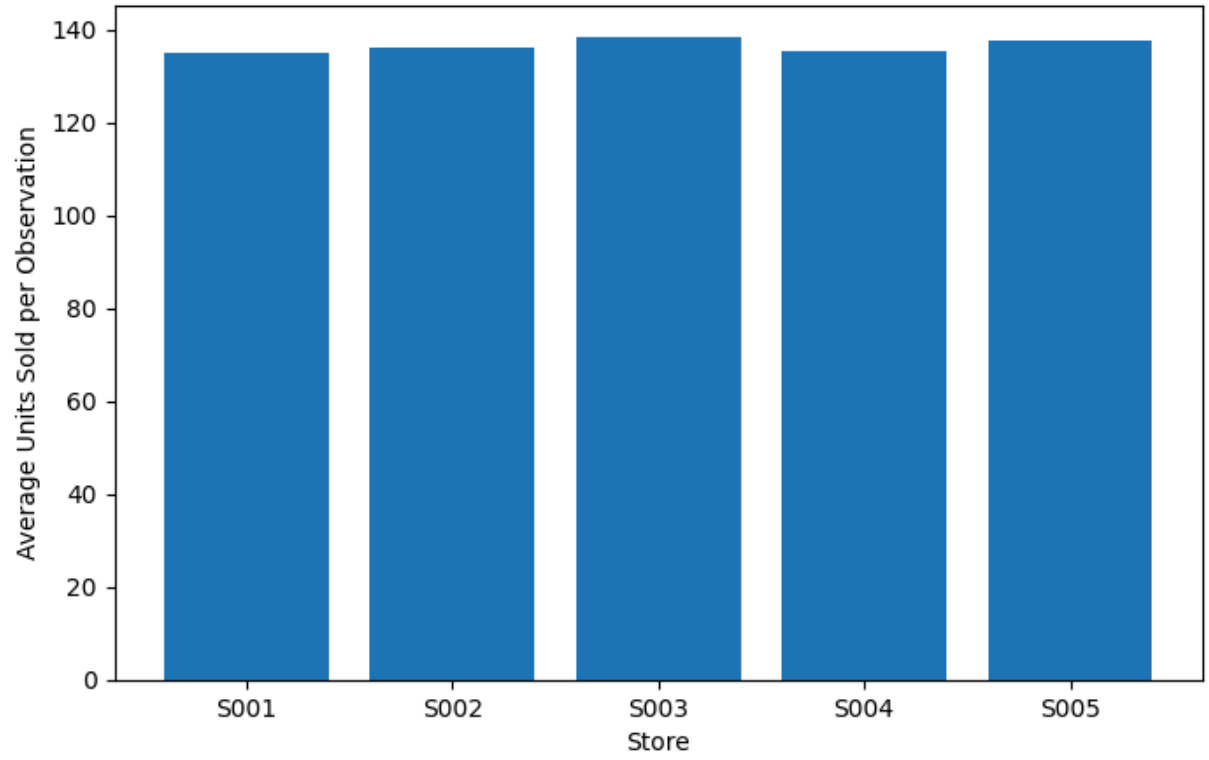
3.3 Store and Product Variation

Sales were relatively consistent across both stores and products.

- Store-level average sales varied only modestly.
- Product-level averages showed similarly limited variation.

This indicated that no single store or product dominated aggregate demand.

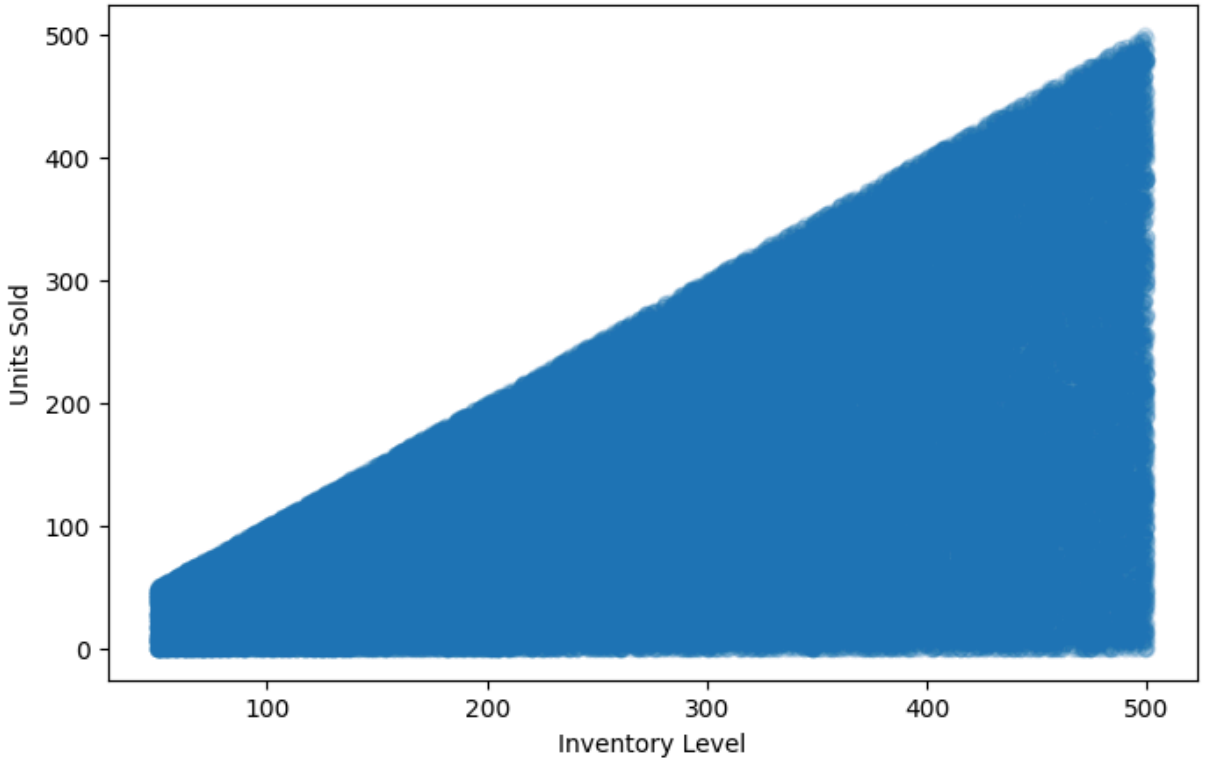
Figure 3: Average Units Sold by Store



3.4 Inventory vs Sales

Inventory Level showed a moderate positive relationship with Units Sold (correlation ≈ 0.59). Higher inventory levels were generally associated with higher observed sales, although this relationship should be interpreted as an association rather than a causal effect.

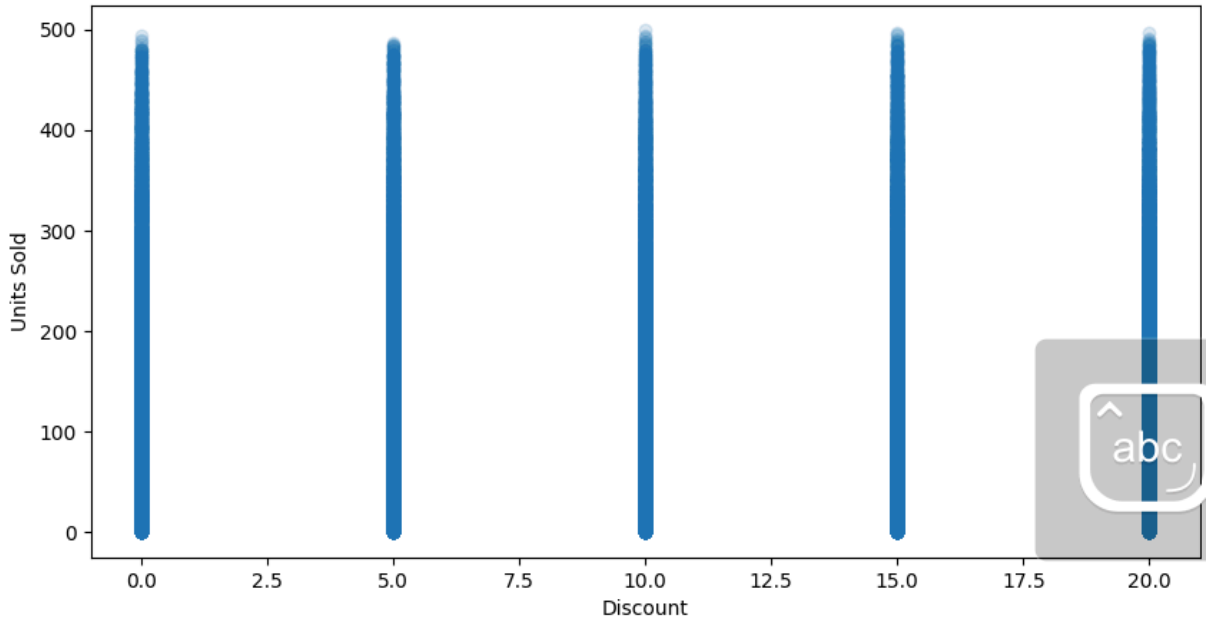
Figure 4: Inventory Level vs Units Sold



3.5 Price/Discount Analysis

Price showed negligible correlation with Units Sold. Discount levels also showed virtually no relationship with sales, with average sales remaining broadly similar across discount levels.

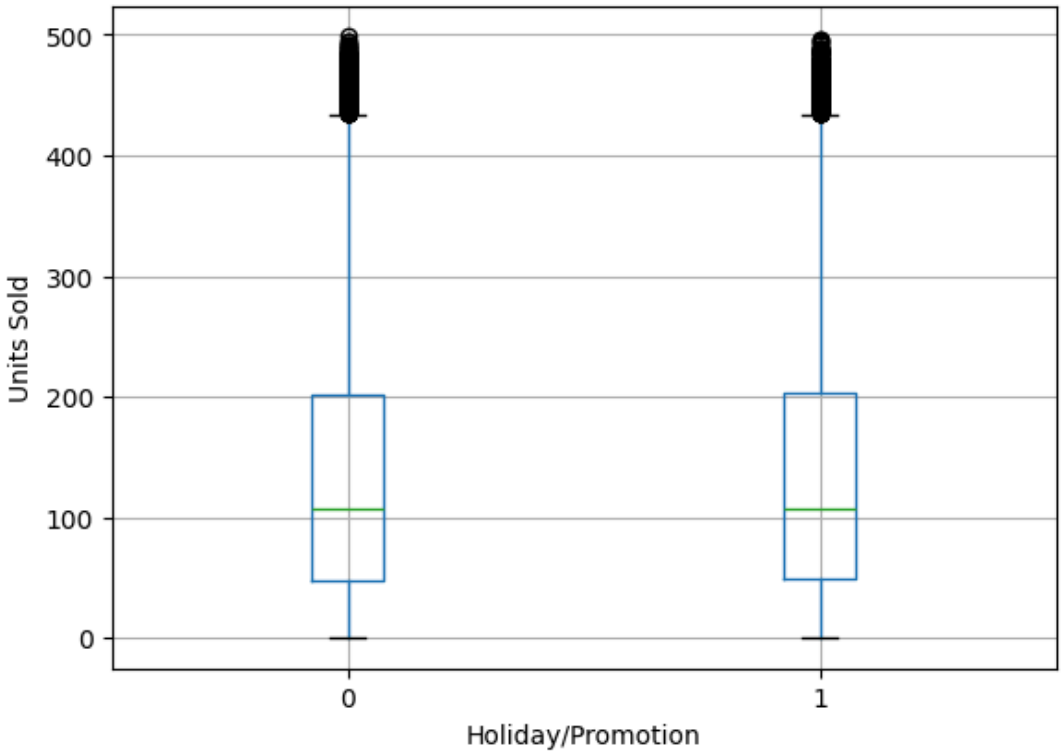
Figure 5: Discount vs Units Sold



3.6 Promotion/Holiday Analysis

The Holiday/Promotion variable showed negligible correlation with Units Sold (approximately -0.004). Average sales were also almost identical between observations with and without the flag.

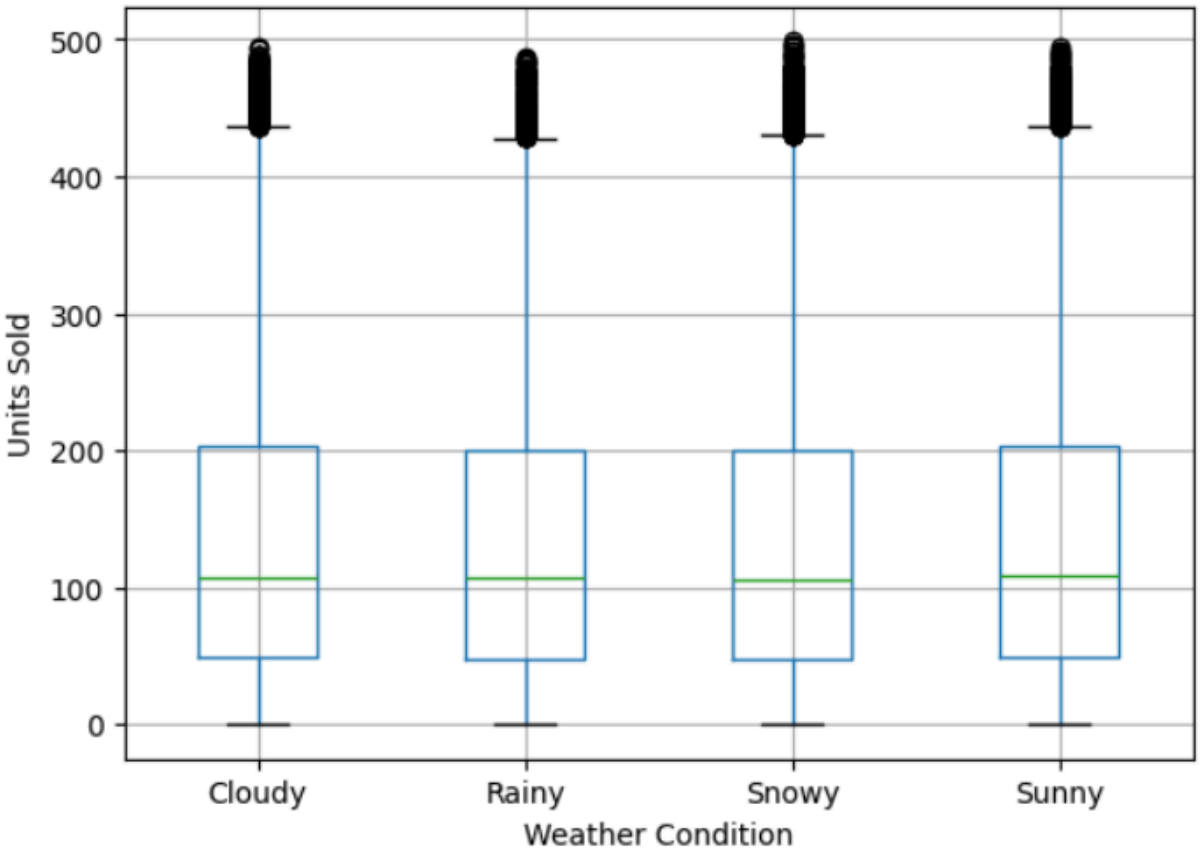
Figure 6: Units Sold by Holiday/Promotion



3.7 Weather and Seasonality

Weather categories showed only small differences in average sales. The supplied seasonality categories similarly showed limited variation in average demand, indicating relatively weak seasonal effects in the dataset.

Figure 7: Units Sold by Weather Condition



3.8 Correlation Analysis

The correlation analysis confirmed that Inventory Level was the most notable relationship with Units Sold, while most other numerical variables showed weak or negligible linear relationships.

Figure 8: Correlation Matrix of Numerical Variables



The Demand Forecast variable showed an almost perfect correlation with Units Sold and was therefore excluded from the primary forecasting models to avoid using an existing forecast as a predictor.

4. Time-Series Analysis

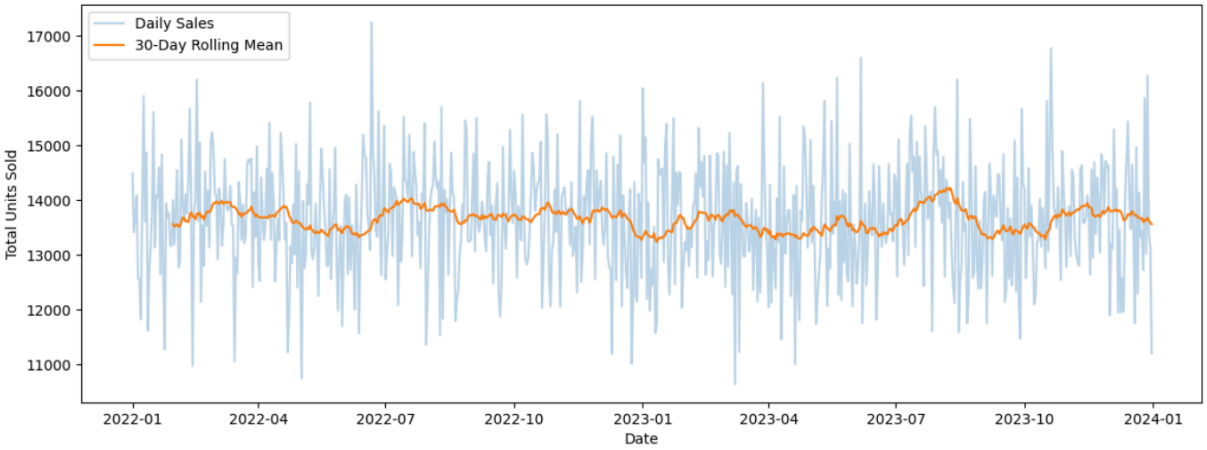
4.1 Daily Target Construction

For forecasting, Units Sold was aggregated across all stores and products to create a daily aggregate sales series.
The time series contained 730 daily observations, covering 1 January 2022 to 31 December 2023.

4.2 Rolling Statistics

A 30-day rolling mean was calculated to assess longer-term movement in daily demand.
The rolling mean remained relatively stable over the analysis period, with no pronounced sustained upward or downward trend.

Figure 9: Daily Sales with 30-Day Rolling Mean



4.3 Train/Test Split

The time series was divided chronologically using an 80/20 split:

- Training set: 584 observations
- Test set: 146 observations

The chronological ordering was preserved to ensure that future observations were not used during model training.

4.4 ACF/PACF

Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were examined to identify temporal dependence in the sales series.

Figure 10: Autocorrelation of Daily Units Sold

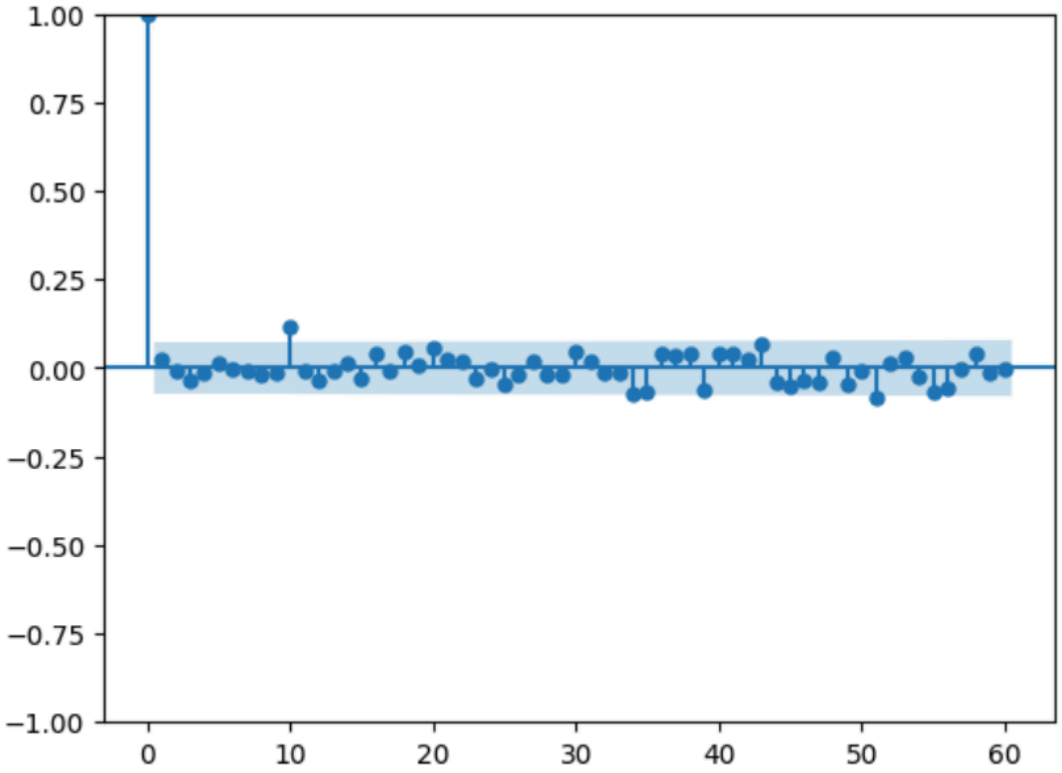
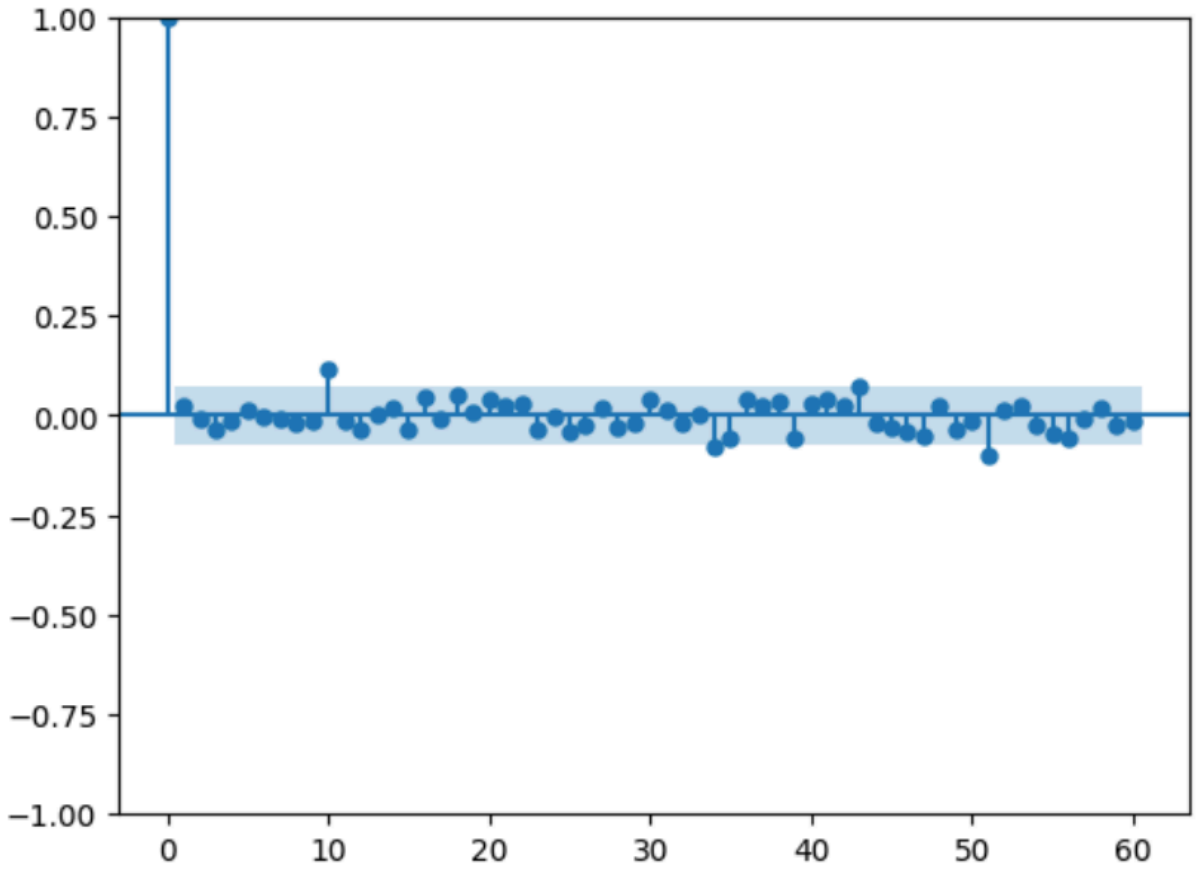


Figure 11: Partial Autocorrelation of Daily Units Sold



Both showed very weak lag dependence, with no prominent lag structure. In particular, there was no strong evidence of a weekly lag pattern.

4.5 ADF Test

The Augmented Dickey-Fuller (ADF) test was applied to assess whether the daily sales series was stationary. The test produced an ADF statistic of approximately -26.29, with a p-value effectively below the numerical precision of the output.

The null hypothesis of a unit root was therefore rejected, providing strong evidence that the series was stationary.

4.6 Interpretation of Stationarity

The stationarity result indicated that the aggregate sales series did not require differencing before modelling.

Consequently:
d = 0
was used in the ARIMA models.

Combined with the weak ACF/PACF structure, this suggested that the historical sales series contained limited temporal dependence, providing the rationale for subsequently testing whether external variables could improve forecasting performance.

5. Forecasting Models

The forecasting stage began with a simple baseline and progressively introduced models capable of capturing autoregressive, seasonal and external-variable effects. Because the time-series diagnostics showed weak temporal dependence, particular attention was given to whether external variables could improve forecasting performance.

5.1 Naive Baseline

A naive forecasting model was used as the benchmark, with each forecast equal to the previous day's observed sales.

The model achieved:

- MAE: 1,171.45
- RMSE: 1,479.49

This provided a reference point for evaluating the more advanced forecasting models.

5.2 ARIMA

ARIMA models were evaluated using the aggregate daily sales series. The ADF test indicated that the series was stationary, so no differencing was required (d=0). Several low-order ARIMA specifications were compared using AIC, BIC and holdout performance.

The simplest model, ARIMA(0,0,0), produced:

- MAE: 845.69
- RMSE: 1,054.17

This was a substantial improvement over the naive baseline. However, the model's forecasts were largely flat because the historical sales series contained very little autocorrelation.

5.3 SARIMA

A seasonal ARIMA model with a weekly seasonal component was tested using SARIMA(0,0,0)(1,0,0,7).

The seasonal component did not provide a meaningful improvement:

- MAE: 1,152.64
- RMSE: 1,441.97

Given the weak ACF at seasonal lags, the results indicated that a weekly seasonal structure was not sufficiently strong to improve the forecast.

5.4 Prophet

Prophet was also evaluated as an alternative forecasting approach, with weekly seasonality enabled.

Performance on the same holdout period was:

- MAE: 875.36
- RMSE: 1,085.78

Prophet performed better than the naive baseline but did not outperform the simpler ARIMA model. It was therefore not selected as the final model.

5.5 SARIMAX

Given the limited temporal structure identified during time-series analysis, the next step was to incorporate external variables into the forecasting framework.

SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous variables) extends ARIMA by allowing additional predictors to influence the target series.
For this project, Units Sold was used as the target, while the following variables were incorporated as exogenous predictors:

- Inventory Level
- Units Ordered
- Price
- Discount
- Holiday/Promotion
- Price Difference
- Weather
- Seasonality
- Region

Preparation of Exogenous Variables

The original dataset contained multiple store-product observations for each date, while the forecasting target was an aggregate daily sales series.

Therefore, the external variables were transformed to the daily level:

- Numerical variables were represented using daily averages.
- Holiday/Promotion was represented as the daily proportion of observations carrying the flag.

- Weather, Seasonality and Region were represented using daily category proportions rather than arbitrarily selecting a single category for each date.
- Categorical variables were converted into numerical features using one-hot encoding, with reference categories omitted.

This produced a modelling dataset with 730 daily observations and 15 exogenous predictors, aligned with the daily sales series.

Model Specification

The initial SARIMAX specification was SARIMAX(0,0,0) with exogenous variables.

This was kept simple because the ACF/PACF analysis had shown little evidence of autoregressive or moving-average structure.

The model was trained on the first 584 observations (80%) and evaluated on the remaining 146 observations (20%), preserving chronological order.

Results

The SARIMAX model achieved:

Table 3: Forecasting Model Performance

Model	MAE	RMSE
Naive	1171.45	1479.49
ARIMA(0,0,0)	845.69	1054.17
SARIMA(0,0,0)(1,0,0,7)	1152.64	1441.97
Prophet	875.36	1085.78
SARIMAX(0,0,0)	648.14	826.16

Compared with the naive baseline, SARIMAX reduced:
- MAE by approximately 44.7%
- RMSE by approximately 44.2%

Compared with the aggregate ARIMA model, it reduced:
- MAE by approximately 23.4%
- RMSE by approximately 21.6%

The forecasts also captured substantially more of the variation in daily sales than the largely flat ARIMA forecasts.

Exogenous Variable Results

The model coefficients indicated that Inventory Level was the strongest statistically significant predictor in the fitted model ($p < 0.001$).

The other variables generally showed weaker statistical evidence. A regional component representing the South category was also statistically significant ($p \approx 0.03$), while the remaining predictors did not show comparable significance at the 5% level.

It is also worth noting that the relationship between inventory and sales is subject to the constraint that observed sales cannot exceed available inventory.

Testing an Autoregressive Component

An additional SARIMAX(1,0,0) specification was tested to determine whether adding an AR(1) term could improve the model.
Its performance was essentially unchanged:

- MAE: 648.53
- RMSE: 826.10
- MAPE: 4.81%

The AR(1) coefficient was also not statistically significant at the 5% level ($p \approx 0.064$).
This indicated that the forecasting improvement was primarily coming from the exogenous variables rather than additional autoregressive structure.

5.6 Why the Final Model Was Selected

The final model was SARIMAX(0,0,0) with exogenous variables.

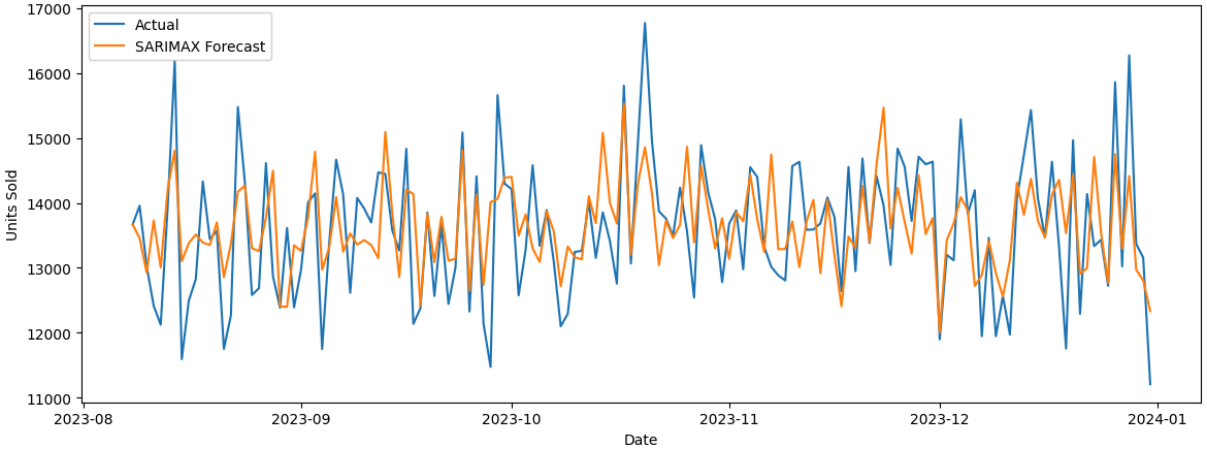
The selection was based on the out-of-sample results:

1. ARIMA established that the aggregate series could be forecast reasonably well despite its weak autocorrelation.
2. SARIMA did not benefit from the introduction of weekly seasonality.
3. Prophet performed better than the naive baseline but did not outperform ARIMA.
4. SARIMAX produced the lowest MAE and RMSE among the tested models.
5. Adding an autoregressive term to SARIMAX produced negligible improvement, so the simpler specification was retained.

The results therefore suggest that, for this dataset, external operational variables contained more useful predictive information than the historical sales series itself.

Final model: SARIMAX(0,0,0) + exogenous variables
MAE: 648.14
RMSE: 826.16
MAPE: 4.81%

Figure 12: SARIMAX Actual vs Forecast



6. Business Insights and Recommendations

The modelling results provide several practical insights for retail demand planning. These recommendations are based on the relationships observed in the dataset.

6.1 Inventory Planning

Inventory Level was the strongest predictor of observed sales in the SARIMAX model ($p < 0.001$), and it also showed a moderate positive correlation with Units Sold (0.59).
This suggests that inventory availability is an important variable to incorporate into demand forecasting and replenishment planning.

Recommendation:
Use inventory levels as a key input when planning replenishment and forecasting demand. Although higher inventory should not automatically be interpreted as causing higher sales, the relationship is quite clear. One possible explanation is that products that are consistently available have more opportunity to be purchased.

Where appropriate, raising the minimum inventory threshold could maintain product availability and reduce situations where limited stock constrains potential sales.

6.2 Pricing

Price showed negligible correlation with Units Sold, while the price difference between the business and its competitor also showed little measurable relationship with demand.

Recommendation:
There is no clear relationship between price, or the difference between the business's price and competitors' prices, and Units Sold in this dataset.

6.3 Discounts

Discount levels showed virtually no relationship with Units Sold in the dataset. Average sales remained broadly similar across the different discount levels examined.

Recommendation:
Routine discounting could potentially be reduced or stopped where its primary purpose is to generate additional demand. The analysis does not indicate a meaningful sales uplift from discounts in this dataset. Discounts should instead be retained for specific commercial objectives and evaluated against incremental sales to determine whether they provide measurable benefits.

6.4 Promotions

The Holiday/Promotion variable showed almost no relationship with Units Sold, with a correlation of approximately -0.004 and nearly identical average sales between flagged and non-flagged observations.

Recommendation:
Promotional and holiday activity should not be considered as a demand driver based on this dataset. Campaign-level analysis would be required to determine whether particular promotions generate incremental sales.

6.5 Regional Considerations

The exploratory analysis did not identify strong differences in average sales across regions. Regional variations were relatively limited compared with the overall variation in daily sales.

Although the South region component was statistically significant in the SARIMAX model ($p = 0.03$), this result should be considered alongside the broader regional analysis rather than treated as evidence of a major regional demand effect.

Recommendation:
There is currently insufficient evidence to justify substantially different inventory or pricing strategies by region based on this dataset alone. Regional segmentation could become more useful if additional information such as store-level characteristics, customer demographics, local demand patterns or geographic factors were available.

6.6 External-Variable Forecasting

The most important modelling result was that incorporating external variables substantially improved forecasting performance.

The final SARIMAX model achieved:

- MAE: 648.14
- RMSE: 826.16
- MAPE: 4.81%

This was considerably better than both the naive baseline and the univariate ARIMA model.

Recommendation:
Retail forecasting should incorporate relevant operational and external information rather than relying exclusively on historical sales. In particular, inventory, pricing, promotions and other variables that are available for the future should be considered when generating demand forecasts.

A practical limitation is that SARIMAX requires future values of its exogenous variables. Therefore, operational deployment would require planned or forecasted values for variables such as inventory, pricing, promotions and other relevant inputs.

Overall Business Implication

The analysis suggests that the greatest practical value of the forecasting approach lies in integrating demand forecasting with inventory rather than treating sales history as an isolated time series. The model can provide a quantitative basis for short-term planning while allowing businesses to incorporate information about their operating environment.

7. Conclusion

This project evaluated the use of time-series forecasting techniques to predict daily retail sales and identify the operational factors associated with changes in demand. The analysis found that the aggregate sales series had limited autocorrelation and weak seasonal structure, which restricted the effectiveness of purely univariate forecasting approaches. ARIMA provided a useful improvement over the naive baseline, while SARIMA and Prophet did not provide sufficient additional benefit.

The introduction of external variables through SARIMAX substantially improved forecasting performance. The final model achieved a MAE of 648.14, RMSE of 826.16 and MAPE of 4.81% on the 20% out-of-sample test period. Adding an autoregressive component produced negligible improvement, indicating that the predictive value came primarily from the external variables rather than additional dependence on previous sales.

From a business perspective, Inventory Level emerged as the most important predictive variable, suggesting that product availability plays an important role in observed sales. In contrast, pricing, discounts, promotions, weather and regional factors showed relatively limited relationships with demand in this dataset.

These findings support using inventory information as an important component of demand and replenishment planning, while reducing reliance on routine discounting where no measurable sales benefit is evident.

Overall, the project demonstrates that retail demand forecasting can be strengthened by combining historical sales with relevant operational variables. The resulting SARIMAX framework provides a practical basis for short-term demand forecasting and can support more informed inventory and supply-chain planning when reliable future values of the external variables are available.